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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fourth Semester B.E. (AI & ML) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21AM404 / 20AM403 / 21CC404 / 20CC503 – OPERATING SYSTEMS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

Unit – I

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 1. a) Discuss with a diagram how context switching happens? | 6 | L*1 | 1 | 1 |
| b) Differentiate the monolithic and microkernel operating systems. | 4 | L2 | 1 | 1 |
| c) Give the importance of various schedulers and queues. | 10 | L2 | 1 | 1 |
| 2. a) What is the difference between threads and processes? Discuss various threading models with neat diagrams. | 6 | L1 | 1 | 1 |
| b) Discuss the states of process with the help of a state diagram. | 4 | L1 | 1 | 1 |
| c) What are the challenges faced for thread scheduling in programming for multicore systems? | 10 | L2 | 1 | 1 |
| 3. a) Discuss the optimization criteria for scheduling algorithm. | 10 | L2 | 2 | 1 |
| b) Consider the following processes arrived at 0ms. | | | | |

Process	Burst time	Arrival time
P1	6	0
P2	8	6
P3	2	2
P4	3	8
P5	5	4

- i) Draw the Gantt chart for SJF, SRTN, FCFS scheduling. What is the turnaround time and waiting time of each process?
- ii) Which algorithm results in minimum average waiting time?

10 L3 2 1

Unit – II

- | | | | | |
|--|---|----|---|---|
| 4. a) Discuss how hardware memory protection is provided using base and limit registers? | 5 | L2 | 3 | 1 |
| b) How paging is implemented using hardware? | 6 | L2 | 3 | 1 |
| c) Explain with diagram the two level directory structure. | 4 | L1 | 3 | 1 |
| d) Consider the following page reference string:
1 0 7 1 0 2 1 2 3 0 3 2 4 0 3 0 2 1 0 7
How many page fault would occur for the FIFO and Optimal page replacement algorithms? Evaluate for 3 page frames. | 5 | L3 | 3 | 2 |
| 5. a) Workout FIFO, LRU page replacement algorithm with 3 frames on the following reference string and give number of page faults and miss ratios: 1,2,3,4,5,6,7,8,9,9,8,7,6,5,4,3,2,1. | 6 | L3 | 3 | 2 |
| b) Explain the indexed allocation of file implementation. | 6 | L2 | 3 | 1 |
| c) With suitable example explain First Fit, Best Fit and Worst Fit memory allocation schemes. Find internal and external fragmentation for the example used. | 8 | L3 | 3 | 1 |

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|----|----|---|----|----|---|---|
| 6. | a) | Write short notes on i) SAN ii) RAID | 10 | L2 | 4 | 1 |
| | b) | Using SCAN disk scheduling algorithm schedule the following disk requests: 23, 168, 56, 122, 44, 33, 98, 101
Initial position is 54. | 6 | L2 | 4 | 1 |
| | c) | Discuss working of moving-head magnetic disk mechanism. | 4 | L1 | 4 | 1 |

Unit – III

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|----|----|---|----|----|---|---|
| 7. | a) | Explain with example Mandatory Access Control and role based access control provides secure access to system resources? | 10 | L2 | 5 | 1 |
| | b) | Discuss how auditing, accounting, logging, and firewall provide security to computer system? | 10 | L1 | 5 | 1 |
| 8. | a) | Summarize how the following security threats effect the system:
i) Revocation of Access Rights ii) trojan horse iii) DoS attack. | 10 | L2 | 5 | 1 |
| | b) | Explain code injection with example. | 10 | L1 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
